Scale-Aware Pruning Framework for Remote Sensing Object Detection via Multifeature Representation

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***Abstract*— With the rapid advancements in computer vision, high-resolution remote sensing imagery has become a crucial data source for object detection. Nevertheless, effectively utilizing limited computational resources and reducing the burden on satellite edge devices remains a significant challenge. To effectively reduce model complexity while maintaining its representational capacity, this article proposes a scale-aware pruning framework (SAPF) to enhance remote sensing object detection ability. First, this article classifies the convolutional layers in object detection models into two categories: layers with a single-scale feature representation and layers with a multiscale feature representation. For convolutional layers with single-scale features, we utilize singular value decomposition (SVD) to quantify feature importance and assess filter redundancy to enhance model efficiency. By removing less critical filters, this pruning criteria aims to reduce the model size and computational load without compromising performance. However, convolutional layers with multiscale features are crucial for optimizing feature extraction and balancing information capture across various scales. To address this, this article evaluates the similarity between convolutional layers with different scales to determine the contribution of various scale features in multiscale fusion. Surprisingly, the SAPF can reduce the FLOPs and parameters, as well as ensure the representational ability obviously when the YOLO v5s and Faster-RCNN are adopted to classify the NWPU VHR-10, RSOD, and SIMD datasets. This means we can save the training computation resources for the model. Additionally, SAPF can significantly improve the efficiency of the model in object detection to ensure its real-time performance.**

***Index Terms*— Multifeature representation, object detection, remote sensing, scale-aware pruning.**

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I. Introduction

OBJECT detection is a pivotal technology in the realm of remote sensing image analysis, specializing in the autonomous recognition and precise localization of specific objects or targets in remote sensing imagery, such as images captured by satellites or drones. In recent years, with the rapid development of convolutional neural networks (CNNs) [1], [2], remote sensing object detection technology has significantly enhanced the ability to extract essential information from large geographic areas. It has been widely applied in fields such as environmental monitoring [3], urban planning [4], traffic management [5], military reconnaissance, resource exploration [6], and scientific research.

Nevertheless, the existing deep learning-based algorithms rely heavily on a vast amount of high-resolution remote sensing images [7], which inevitably leads to scenarios where some deep learning models encounter previously unseen objects in practical applications. Consequently, these manually designed complex networks for object detection need to be retrained to incorporate new knowledge [8]. However, the retraining process is time-consuming and resource-intensive, which can strain the hardware of satellite edge devices. In addition, with the continuous increase in the resolution and volume of remote sensing images, remote sensing object detection tasks require significant computational resources and storage space. Given the limited computational capacity available on satellites, improving the computational efficiency and optimizing resource utilization of object detection models is an urgent problem that needs to be solved. To address these challenges, it is crucial to employ network compression techniques to optimize model structures and conserve computational resources.

Currently, pruning techniques [9], [10], [11] are the most popular method among network compression techniques applied to object detection, which aims to reduce model complexity and save computational resources by removing unimportant connections or parameters. In the literature, common network pruning methods include channel pruning (CP), weight pruning, and dynamic pruning (DP). Pruning methods

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for object detection can be broadly classified into two main categories: coarse-grained pruning [12], [13], [14] and fine-grained pruning [15], [16], [17]. Fine-grained pruning typically involves removing individual weights from any position within the weight tensor, which makes the network structure sparse to reduce computational overhead. However, the irregularity of the sparse structure resulting from fine-grained pruning poses challenges for current hardware architectures, which prevents the acceleration of these models effectively. Consequently, the computational efficiency of networks after fine-grained pruning remains a significant challenge. In contrast, coarse-grained pruning involves removing weights at the granularity of entire channels or convolutional kernels, resulting in a more regular network structure. While this regularity aids in model acceleration, coarse-grained pruning may not effectively optimize multiscale convolutional layers based on information from different scales, potentially leading to the loss of critical feature information in object detection.

To overcome the limitations of traditional pruning methods, this article proposes a scale-aware pruning framework (SAPF) that leverages multifeature representation to reduce the complexity of object detection models and alleviate the burden on edge devices. Given that convolutional layers in object detection models vary in feature extraction capabilities depending on their scale, we categorize the convolutional layers of the object detection model into single-scale convolutional layers and multiscale convolutional layers based on scale information. Single-scale convolutional layers handle feature maps with a uniform spatial scale. Due to their simpler structure and fixed feature scale, pruning strategies focus more on evaluating the local importance of features. In contrast, multiscale convolutional layers handle feature maps from various scales, involving different resolutions and information levels. Since feature maps at different scales contribute differently and have distinct integration needs, pruning strategies must consider the different impacts of feature maps from various scales on the final object detection. To ensure precise pruning, this article proposes two distinct criteria for single-scale and multiscale convolutional layers. For single-scale convolutional layers, this article uses singular value decomposition (SVD) to map the weight matrix of convolutional kernels from the weight space to the feature space. The magnitude of the singular values indicates the extent to which a filter reduces the uncertainty in the output of the model, thereby assessing filter redundancy. Larger singular values signify that the filter significantly contributes to the feature representation of model, meaning it is more important. Conversely, smaller singular values suggest that the filter is less critical and can be removed. By eliminating less important filters in the weight matrix, the complexity and computational load of the model can be substantially reduced.

Multiscale convolutional layers typically perform convolution operations on feature maps at various scales to capture information across different resolutions. Considering that feature maps from different scales in the multiscale fusion module of an object detection model may contribute differently to the object detection task, this article designs a scale importance score (SIS) as a pruning indicator for multiscale convolutional layers. The SIS reflects the information richness of feature maps at different scales and facilitates DP of multiscale convolutional layers. By evaluating the variation and information richness of the feature maps output by different-scale convolutional layers, we identify the most information-rich convolutional layers. This layer serves as a benchmark for comparing other layers, allowing us to determine the SIS for each multiscale convolutional layer. A lower SIS value indicates higher information redundancy, which helps in effectively and precisely identifying which layers to prune and the extent of pruning required. Through these targeted pruning operations, we can obtain a pruned model for remote sensing object detection that maintains strong representational capability while reducing complexity.

Three contributions are summarized in this article.

1) This article introduces an SAPF for object detection, which leverages feature representation. SAPF employs two distinct pruning criteria tailored for single-scale and multiscale convolutional layers. This framework not only preserves the accuracy of remote sensing object detection but also optimizes computational resource usage, thereby enhancing the training efficiency of the object detection model.
2) For the multiscale fusion module in object detection models, this article introduces the SIS metric to achieve precise pruning of multiscale convolutional layers. The SIS metric assesses the redundancy of these layers based on the information richness at different scales, facilitating more effective and targeted model pruning.
3) Impressive pruning results are accomplished on Faster-RCNN [18] and YOLO v5s [19] with NWPU VHR-10 [20], RSOD [21], SIMD [22], including the reductions of FLOPs and parameters, as well as the improvement of accuracy.

This article is structured as follows. In Section II, we introduce the related work of remote sensing object detection, as well as the analysis of traditional pruning methods. In Section III, we introduce and explain our SAPF for object detection in detail. In Section IV, we verify our method on different datasets and models, as well as make an analysis of related results. Finally, the conclusion of the whole work is drawn in Section V.

II. Related Work

In this section, we divide the content into two parts. First, this article introduces the background knowledge of remote sensing object detection and related deep learning techniques. Finally, we introduce the existing mainstream traditional network pruning techniques and analyze their advantages and disadvantages.

A. Remote Sensing Object Detection Technologies

Recently, deep learning techniques have rapidly advanced remote sensing object detection. As the resolution and volume of satellite images increase, there is a need for advanced methods that can detect and classify objects with high accuracy. Deep learning models, especially CNNs [23], [24], [25], [26], excel at learning complex features from large datasets,

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which is crucial for analyzing diverse objects in remote sensing imagery. Unlike traditional methods that rely on handcrafted features, deep learning offers significant improvements in detection accuracy and robustness.

Several notable deep learning-based approaches have been developed to address the challenges of remote sensing object detection. Deng et al. [27] designed a multiscale feature fusion module to address the challenge of detecting small objects in remote sensing images. This method extracts features at different scales and combines them with global context information, significantly improving the detection accuracy of small objects. Liu et al. [28] introduced a new region proposal network (RPN) structure that enhances the ability of the model to detect objects in dense and complex backgrounds. The improved Faster-RCNN algorithm better identifies objects and reduces false detection rates when processing high-resolution remote sensing images. Zhang et al. [29] proposed a new feature fusion mechanism that effectively integrates feature information from different scales, thereby enhancing object detection capabilities. This method is particularly suitable for handling complex objects in high-resolution remote sensing images and has demonstrated excellent detection performance on standard datasets.

While numerous deep learning techniques have been developed for remote sensing object detection, addressing the challenge of limited computational resources remains an ongoing issue. Although advancements in model architectures and optimization strategies have significantly improved detection performance, these methods often require substantial computational power and memory, which can be a limitation in resource-constrained environments such as satellite systems. The need for efficient algorithms that balance detection accuracy with computational feasibility is crucial for deploying deep learning models in practical remote sensing applications, where computational resources are limited. Further research is needed to develop lightweight and efficient models that can operate effectively under such constraints without compromising performance.

B. Existing Mainstream Traditional Network Pruning Techniques

In existing work, pruning techniques are mainly categorized into four types: CP, block pruning (BP), weight cascading pruning (WCP), and DP. WCP primarily involves iteratively pruning and retraining to progressively optimize the model, aiming for a significant reduction in parameter count while maintaining and enhancing model performance. Han et al. [30] proposed a method that simultaneously learns weights and connections, achieving efficient neural network models through weight pruning and iterative optimization. Li et al. [31] designed a filter-level pruning method that dynamically selects and prunes filters to compress CNNs. DP is a method that adjusts the network structure dynamically based on real-time needs during training or inference. Its core idea is to dynamically increase or decrease the structure of the network and parameters based on current input data or intermediate results during inference, optimizing the performance and efficiency of the model under varying conditions. Meng et al. [32] proposed a method using an adaptive vision transformer (AdaViT) for efficient object detection. The author proposes an adaptive computation framework that dynamically adjusts the number of patches, attention heads, and Transformer blocks used during inference based on the complexity of the input image to reduce unnecessary computations. Additionally, AdaViT inserts a lightweight decision network before each Transformer block. This network is responsible for predicting which patches, attention heads, and Transformer blocks should be activated or skipped. This dynamic decision-making mechanism is implemented in an end-to-end trainable manner using Gumbel-Softmax technology. Luo and Wu [33] proposed an end-to-end trainable filter pruning method aimed at improving the efficiency of deep model inference. However, despite their flexibility, WCP and DP have notable challenges and limitations in terms of sparsity management, computational efficiency, and fine-tuning requirements.

Compared to unstructured pruning, CP and BP can more effectively reduce the number of parameters and optimize the use of computational resources by removing entire structural units. CP reduces computational load and storage requirements by selectively removing less important channels in each convolutional layer based on channel importance evaluation. He et al. [34] proposed a CP method based on geometric median, which accelerates deep CNNs by removing less important filters according to their importance ranking and geometric median. Liu et al. [35] introduced a network slimming framework that optimizes binary masks for each channel during training to selectively retain important channels, while promoting channel sparsity through L1-norm regularization. BP involves pruning convolutional kernels in blocks to maintain the structure of the kernels and the regularity of the model.

III. Methodology

A. Overview

In this section, the specific implementation of the proposed method will be described in detail. The overall structure of the SAPF framework is illustrated in Fig. 1. On the left side of Fig. 1, the original images for object detection are provided and labeled. In object detection models, convolutional layers of different scales typically play different roles in feature extraction and task execution. Based on this, this article proposes an SAPF. SAPF aims to divide the convolutional layers in object detection models into single-scale convolutional layers and multiscale convolutional layers by determining whether the number of scaling factors in the convolutional layers is unique during model training, i.e., the convolutional layer with a unique scaling factor is identified as a single-scale convolutional layer; the convolutional layer containing multiple scaling factors is considered a multiscale convolutional layer. The upper portion of Fig. 1 depicts the multiscale convolutional layers and the corresponding pruning criteria. The pruning strategy involves using convolutional layers at different scales as units to identify the most information-rich convolutional layer, which serves as a reference for measuring the redundancy of convolutional layers at other scales. This approach ensures the key information of multiscale features.

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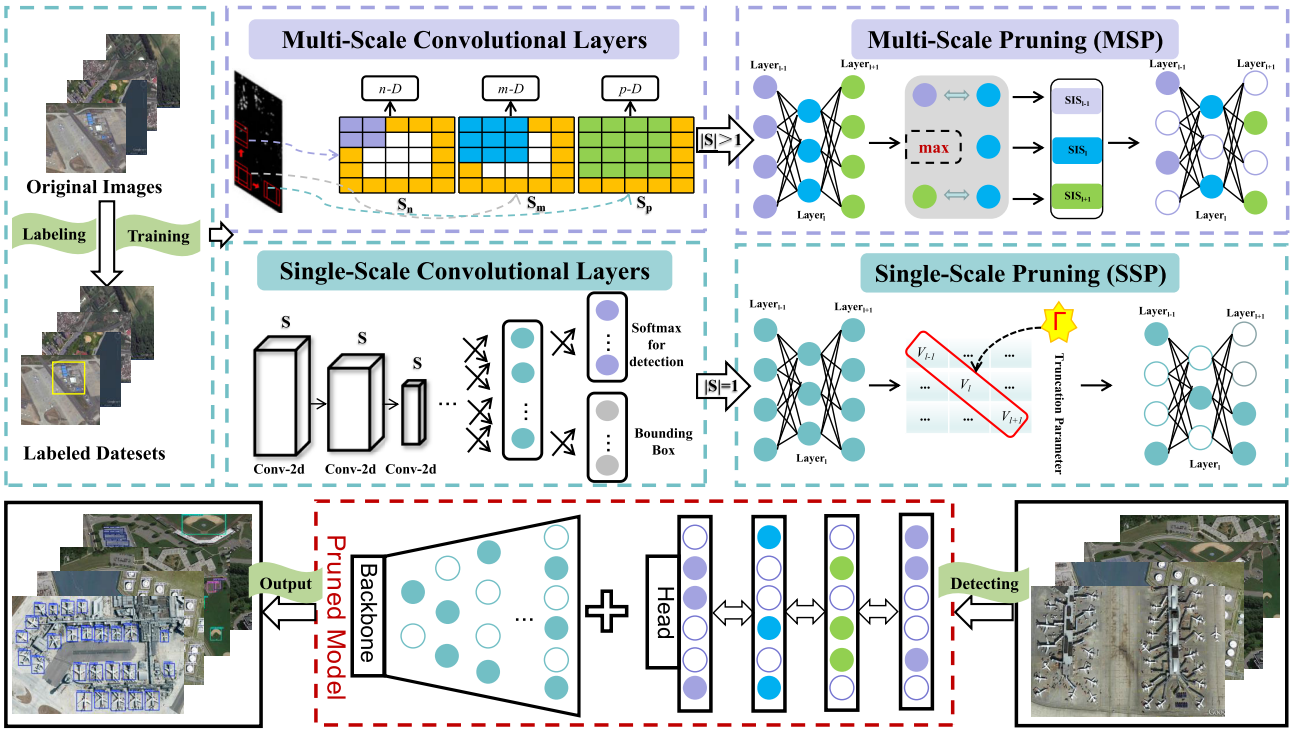


Fig. 1. Flow of SAPF. In the upper left corner is the labeled dataset required for training an object detection model. The convolutional layers of the object detection model are divided into multiscale convolutional layers and single-scale convolutional layers. The top section shows the process of MSP. The middle section illustrates the process of SSP. The bottom section shows the unlabeled dataset required for detection in the object detection model. The detection is performed by the pruned model, which includes both MSP and SSP and finally outputs the detection results. Among them, S, S_n, S_m, and S_p represent different scale factors of the convolutional layer, respectively; $|S|$ represents the number of scales of the convolutional layer; V represents the feature values that the filter maps from the convolutional space to the feature space.

The middle part of Fig. 1 illustrates the single-scale convolutional layers and their corresponding pruning criteria. Given that single-scale convolutional layers have uniform feature map sizes, the importance of filters is measured using the SVD technique. This allows for the precise pruning of redundant filters. Finally, the pruned model is used for object detection to output the final object detection results in the bottom part of Fig. 1 after the single-scale and multiscale pruning (MSP) processes.

B. Scale-Aware Pruning Strategies

In this part, the specific implementations of the two pruning criteria for single-scale convolutional layers and multiscale convolutional layers in SAPF will be detailed. During the training process of the object detection model, the original image (I) is resized to different scales by applying various scaling factors ($\mathbf{S}$), i.e., $I_s = \text{Resize}(I, \mathbf{S})$. For single-scale convolutional layers, the scaling factor is unique because the convolutional kernels are uniform. In contrast, for multiscale convolutional layers, multiple convolutional kernels $\mathbf{K} = \{K_1, K_2, \ldots, K_N\}$ may correspond to multiple scaling factors $\mathbf{S} = \{S_1, S_2, \ldots, S_N\}$. The scaling factor is determined through the following calculation:

$$\begin{aligned}\mathbf{S} &= (S_W, S_H) \\ &= \left(\frac{W_{\text{out}}}{W_{\text{in}}}, \frac{H_{\text{out}}}{H_{\text{in}}}\right) \\ &= \left(\frac{\frac{W_{\text{in}}-K_W+2P}{S}+1}{W_{\text{in}}}, \frac{\frac{H_{\text{in}}-K_H+2P}{S}+1}{H_{\text{in}}}\right)\end{aligned} \tag{1}$$

where $(W_{\text{in}}, H_{\text{in}})$ represents the width and height of the input feature map, respectively; and $(W_{\text{out}}, H_{\text{out}})$ adopts the width and height of the input feature map, respectively. (K_W, K_H) denotes the width and height of the convolutional kernel, respectively. S stands the stride, and P denotes the padding size. Therefore, we can evaluate the pruning criterion (PC) based on whether the scaling factor is unique, as shown in the following relationship:

$$\text{PC} = \begin{cases} \text{SSP}, & |\mathbf{S}_l| = 1 \\ \text{MSP}, & |\mathbf{S}_l| > 1 \end{cases} \tag{2}$$

here, $|\mathbf{S}_l|$ represents the number of scaling factors in the lth convolutional layer. This means that convolutional layers with unique scaling factors use a single-scale pruning (SSP) strategy, while convolutional layers with multiple scaling factors employ an MSP strategy.

In object detection models, the feature maps of single-scale convolutional layers are typically of uniform size. The convolution operation for single-scale is generally as follows:

$$(\mathbf{M}_i)_{(x,y)} = \sum_{i=0}^{K_W-1} \sum_{j=0}^{K_H-1} (C_i)_{(x+i,y+j)} \cdot (K)_{(i,j)} \tag{3}$$

where $(\mathbf{M}_i)_{(x,y)}$ stands for the value of the output feature map $(\mathbf{M}_i)$ at position (x, y). The term $(C_i)_{(x+i,y+j)}$ refers to the

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value of the input feature map C_i at position $(x+i, y+j)$. Meanwhile, $(K)_{(i,j)}$ denotes the value of the convolution kernel K at position (i, j).

Due to the diversity of scaling factors in multiscale convolutional layers, the feature maps generated by convolutional layers at different scales have varying dimensions

$$(\mathbf{M}_i)^n_{(x,y)} = \sum_{i=0}^{K_W^n-1} \sum_{j=0}^{K_H^n-1} (C_i^n)_{(x+i,y+j)} \cdot (K^n)_{(i,j)} \tag{4}$$

where n represents nth scale. Moreover, in multiscale convolutional layers, feature maps at different scales are fused to create richer feature representations, typically as shown below

$$\mathbf{M}_i' = \mathbb{U}(\mathbf{M}_{i+1} + C_i^n) \tag{5}$$

where $\mathbf{M}_i'$ is the feature map after feature fusion, and $\mathbb{U}$ represents the upsampling process.

1) SSP: For single-scale convolutional layers, the SSP process is illustrated in Fig. 2. Since the filters in single-scale convolutional layers usually have fixed sizes and handle features at a single scale, this article uses SVD to map the convolution kernels from the convolutional space (original space) to the feature space (low-rank space). The convolutional space refers to the space where the weight matrices of the convolutional layer reside. In a convolutional layer, the convolution kernels typically exist as high-dimensional matrices. First, we typically perform dimensionality reduction along the two dimensions of the input image: height m and width n, as shown in in the following equations:

$$\tilde{\mathbf{A}}^{\max}_{(i,j,:)} = \max_n\left(\mathbf{A}_{(i,:,n,j)}\right), \quad \tilde{\mathbf{A}}^{\max} \in \mathbb{R}^{i*m*j} \tag{6}$$

$$\mathbf{A}^{\max}_{(i,j)} = \max_m\left(\tilde{\mathbf{A}}^{\max}_{(i,m,j)}\right), \quad \mathbf{A}^{\max} \in \mathbb{R}^{i*j} \tag{7}$$

where (i, j) stands for the input channel and the output channel, respectively. $\mathbf{A}_{(i,:,n,j)}$ represents the slice of width dimension at a specific height position for the ith input channel and jth output channel. $\mathbf{A}^{\max}_{(i,j)}$ denotes the maximum value across all height positions for the ith input channel and jth output channel, which means that folding along the dimensions m and n is done by taking the maximum value.

After reducing the convolutional space to a low-rank space, SVD decomposes the convolution matrix into a product of three matrices, which have lower dimensions in the feature space. Specifically, SVD decomposes the convolution kernel matrix $\mathbf{A}$ into

$$\mathbf{A} = \mathbf{U}\sum\mathbf{V}^T, \quad \mathbf{A} \in \mathbb{R}^{i*m*n*j} \tag{8}$$

where $\mathbf{U} \in \mathbb{R}^{m*n}$ represents left singular vector matrix; $\mathbf{V}^T \in \mathbb{R}^{m*n}$ represents right singular vector matrix; $\sum \in \mathbb{R}^{m*n}$ stands for diagonal singular value matrix, which $\sum = \text{diag}(\sigma_1, \sigma_2, \ldots, \sigma_r)$, with σ_r denotes singular value.

By evaluating the magnitude of the singular values, the importance of the filter can be determined. This means that larger singular values contain more crucial information, indicating lower redundancy. To more conveniently assess whether a filter can be retained, this article introduces a truncation threshold Γ. This means that filters with singular values

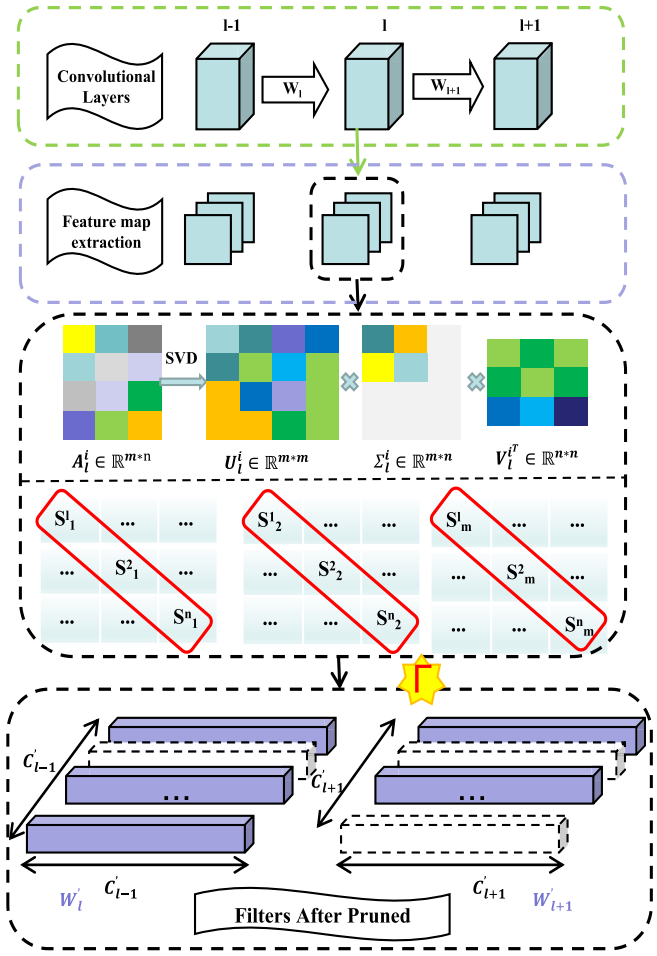


Fig. 2. Process of SSP. w_l stands for the weights of the lth layer. The SVD technique maps filters from the convolutional space to the feature space to assess the importance of the filters, enabling precise pruning.

exceeding Γ should be retained, while those below Γ should be discarded. The relationship between the index of the singular value and the truncation there hold is $I = \{i | \sum(i, i) \geq \Gamma\}$. The diagonal matrix can be reconstructed according to the truncation threshold Γ as follows:

$$\begin{aligned} \tilde{\sum} &= \text{diag}(\sum(i, i) | i \in I) \\ \mathbf{U}' &= \mathbf{U}[:, I] \\ \mathbf{V}' &= \mathbf{V}[:, I]. \end{aligned} \tag{9}$$

2) MSP: For multiscale convolutional layers, the MSP process is illustrated in Fig. 3. In object detection models, multiscale convolutional layers are responsible for capturing features at different scales and levels. Each convolutional layer at different scales has a distinct receptive field and processes different feature information. SVD is typically used for dimensionality reduction in linear transformations, and it may not adequately capture and preserve important features in convolutional layers with multiscale feature information. To address this issue, this article proposes a scale-aware multiscale PC. By measuring the similarity between convolutional layers, it is possible to understand the redundancy among the layers better and make more informed pruning

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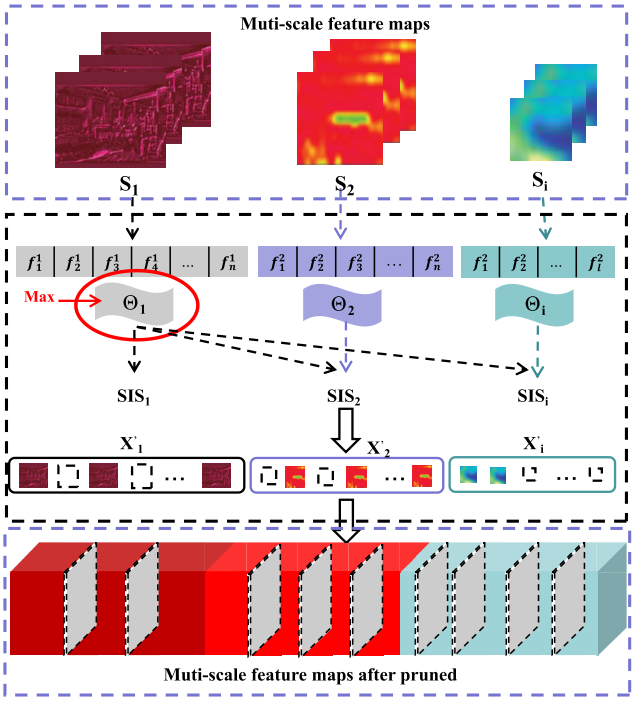


Fig. 3. Process of MSP. S_i stands for the scaling factor for different feature maps, which means the size of the feature maps varies. f_i^n represents the nth filter under the ith scaling factor. θ_i measures the richness of feature information in the ith convolutional layer. SIS denotes the similarity between convolutional layers after comparison, thereby determining the redundancy of filters in each convolutional layer. The final filter structure retained across different scales of convolutional layers is denoted as X_i'.

decisions. Filters in convolutional layers at different scales can be represented as f_i^n, where i denotes the ith scaling factor, and n represents the number of filters. We obtain the richness of feature information for each scale of the convolutional layer by calculating the variance of the filters in that layer

$$\theta_i = \mathrm{VAR}(f_i^n) = \frac{1}{n}\sum_{j=1}^{n}\left(f_{(i,j)} - \overline{f}_i\right)^2 \tag{10}$$

where $f_{(i,j)}$ represents the output of the jth filter at the ith scale, $\overline{f}_i$ denotes the average output of all filters, and θ_i measures the richness of feature information at that scale. By calculating the variance, the stability and the amount of information of the filter outputs at this scale can be assessed.

SIS is designed to measure the similarity between multi-scale convolutional layers to more effectively understand and optimize the feature fusion process. SIS is obtained using the following equation:

$$\mathrm{SIS}_i = \frac{e^{\sum_{i=1}^{n}\theta_i \ln \frac{\theta_i}{\theta_{\max}}}}{\sum_{i=1}^{n} e^{\sum_{i=1}^{n}\theta_i \ln \frac{\theta_i}{\theta_{\max}}}} \tag{11}$$

where $\theta_{\max}$ represents the convolutional layer i with the highest feature richness. After assessing the importance of feature information at different scales using the SIS metric, we can determine the filter retention status for different convolutional layers, as follows:

$$X_i' = X_i(1 - \mathrm{SIS}_i \cdot \beta) \tag{12}$$

Algorithm 1 SAPF

Input: The Number of the model layer L;
Scaling factors **S** for scale-aware convolutional layers;
Learning ratio η;
Truncation threshold Γ;
Global importance sensitivity value β.
Output: ***The pruned model: M***.

```
while l<L do
    if |S_l| = 1 then
        PC = SSP.
        for l < L do
            map the filters of convolutional layer l to the feature space as shown in Eq. (6) - (7);
            Quantify the importance of each eigenvalue using SVD techniques as shown in Eq. (1);
            Remove filters with importance below Γ as shown in Eq. (9).
    if |S_l| > 1 then
        PC = MSP.
        for l < L do
            Calculate the feature representation capability θ_i of convolutional layers at different scales;
            Compare the similarity between multi-scale convolutional layers to derive the SIS_i according Eq. (11);
            Based on Eq. (12), combine SIS_i and β to determine the filter structure X'_i that should be retained.
```

where SIS_i stands for the importance of scale feature information representation for the ith convolutional layer, and β represents the global importance sensitivity value across multiple scales. The filter structure that needs to be retained for the ith scale convolutional layer is obtained through the operations in (11), denoted as X_i'.

C. Implementation of SAPF

In this part, we propose the SAPF to determine the filter structure that should be retained across different scales in convolutional layers. The detailed implementation of SAPF is illustrated in Algorithm 1. The uniqueness of the scaling factor is used as a criterion to distinguish between single-scale and multiscale convolutional layers. Convolutional layers with a unique scaling factor are classified as single-scale layers, while those with diverse scaling factors are classified as multiscale layers. For single-scale convolutional layers, the redundancy of filters is assessed by mapping the filters from the convolutional space to the feature space. For multiscale convolutional layers, SIS is obtained by comparing the feature representation capabilities across different scales to evaluate the richness of feature information among the multiscale convolutional layers. The pruning situation of multiscale convolutional layers is evaluated using SIS. By combining two specific pruning criteria, an object detection model with strong

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feature representation capability and reduced complexity is ultimately obtained.

IV. Experiments

To validate the effectiveness of SAPF, we applied it to Faster-RCNN [18] and YOLO v5s [19] on three different object detection datasets. This section is divided into five parts: the introduction of datasets, experimental details and related metrics ablation, experimental analysis, parameters sensitivity analysis, and comparative experiment analysis.

A. Datasets

In this article, three benchmark remote sensing object detection datasets are employed on the Faster-RCNN [18] and YOLO v5s [19] models to demonstrate the effectiveness of SAPF, i.e., the NWPU VHR-10 [20] datasets, the RSOD [21] datasets, and the SIMD [22] datasets. To ensure a fair comparison, three object detection datasets are divided into training and testing sets at an 8:2 ratio. Additionally, all remote sensing images in the three datasets were resized to a resolution of 640 × 640 pixels.

1) NWPU VHR-10 Datasets [20]: The NWPU VHR-10 Dataset [20] is a high-resolution remote sensing imagery dataset developed by Northwestern Polytechnical University, focusing on object detection and recognition. This dataset includes a variety of high-resolution satellite images covering different scenes such as urban and rural areas, with rich annotation information that provides valuable resources for research in remote sensing image processing and computer vision. The dataset contains a total of 800 images, including 650 annotated images with objects and 150 background images. It features ten object categories: aircraft, ships, oil tanks, baseball fields, tennis courts, basketball courts, athletic fields, ports, bridges, and cars. The NWPU VHR-10 dataset [20] is characterized by its high resolution and detailed annotations, making it highly effective for tasks such as object detection, land cover classification, and change detection.

2) RSOD Datasets [21]: The RSOD dataset [21] is a high-quality resource specifically designed for remote sensing object detection tasks. It is primarily intended to advance the technology of object detection and recognition in remote sensing images. The RSOD dataset [21] includes a variety of high-resolution remote sensing images that cover different types of objects, such as vehicles, buildings, ships, and aircraft. The images in the dataset span various environments, including urban, rural, and industrial areas, making it suitable for a range of application scenarios. These images are typically sourced from various satellite and aerial platforms and come with detailed annotation information, including object bounding boxes and category labels.

3) SIMD Datasets [22]: The SIMD dataset [22] was introduced by Haroon et al. and developed by the National University of Sciences and Technology (NUST) in Pakistan, which is primarily designed for vehicle detection. The dataset includes 5000 high-resolution remote sensing images, each with a size of 1024 × 768 pixels. It contains a total of 45 096 annotated instances, categorized into 15 different classes. Although the dataset is primarily focused on vehicle detection, its diverse instance categories and high-quality annotations make it valuable for various remote sensing object detection tasks.

B. Experimental Details and Related Metrics

To ensure fairness and consistency in the experiments, we deployed the experiments on a PyTorch framework with CUDA VERSION 11.6, using an NVIDIA RTX 2080 Ti graphics card. The model was trained for 100 epochs with a batch size of 16. The learning rate was set to range from 0.01 to 0.001 during training.

In the experiments conducted in this article, we primarily evaluate the performance of the Faster-RCNN [18] and YOLO v5s [19] models on three object detection datasets using four metrics: mAP, efficiency, FLOPs, and parameters. Among these metrics, mAP@50, mAP@75, and mAP@(50:95) represent the mean average precision at IoU thresholds of 0.5, 0.75, and across the range from 0.5 to 0.95, respectively. mAP indicates the ability of the model to represent features before and after pruning. Efficiency demonstrates the performance of the model before and after pruning. FLOPs and parameters provide a clear representation of the structural lightness of the model before and after pruning.

C. Ablation Experimental Results and Analysis

In this part, the ablation experiments will be divided into two main parts: the results and analysis of the ablation experiments for Faster-RCNN [18], and the results and analysis of the ablation experiments for YOLO v5s [19]. The ablation experiments for each model are mainly divided into four parts: fully trained models, SSP, MSP, and SAPF. Among these, the results of the fully trained models serve as baseline results for comparison with other methods. SSP refers to the detection results of the model after pruning only the single-scale convolutional layers. MSP represents the detection results of the model after pruning only the multiscale convolutional layers. SAPF denotes the detection results of the model after complete pruning of all different-scale convolutional layers.

1) Results and Analysis of the Ablation Experiments for Faster-RCNN [18]: The ablation experiment results for Faster-RCNN [18] are shown in Table I. For the NWPU VHR-10 dataset [20], MSP improves accuracy by approximately 2% compared to SSP. This indicates that some filters in the multiscale convolutional layers negatively impact feature representation. MSP effectively identifies and removes these redundant filters, enhancing the feature representation of Faster-RCNN [18] on this dataset. SAPF not only achieves a stronger pruning level than MSP but also maintains superior feature representation capability, demonstrating the effectiveness of SAPF. The RSOD dataset [21] is relatively small, so the accuracy across models is generally similar. However, experimental results indicate that SAPF significantly reduces FLOPs and parameters compared to the original model. This means that the pruned structure obtained with SAPF is more lightweight, conserving computational resources and markedly improving model efficiency. For the SIMD dataset [22], the

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TABLE I
ABLATION EXPERIMENT RESULTS FOR THE FASTER-RCNN [18] MODEL. THE BOLDED PARTS ARE THE SAPF METHOD AND ITS RELATED METRIC RESULTS

Model	Datasets	SSP	MSP	mAP@50 (%)	mAP@75 (%)	FLOPs (G)	Parameters (M)	Efficiency (GPUDays/Epochs)
Faster-RCNN [18]	NWPU VHR-10 [20]			87.40	61.30	64.23	41.39	2.74×10^{-3}
		✓		87.00	54.60	60.40	37.73	2.27×10^{-3}
			✓	89.30	61.90	62.63	41.04	1.73×10^{-3}
		✓	✓	**89.40**	**62.00**	**60.34**	**37.36**	**0.78×10^{-3}**
	RSOD [21]			90.40	69.60	82.98	41.36	2.03×10^{-3}
		✓		90.50	68.70	77.74	37.71	1.89×10^{-3}
			✓	90.10	72.90	80.79	41.01	1.92×10^{-3}
		✓	✓	**90.40**	**68.80**	**75.02**	**37.35**	**1.26×10^{-3}**
	SIMD [22]			92.30	91.20	71.72	41.42	8.54×10^{-3}
		✓		92.20	90.90	67.33	37.76	7.83×10^{-3}
			✓	92.50	91.30	70.703	41.22	7.93×10^{-3}
		✓	✓	**92.10**	**91.70**	**65.52**	**37.43**	**7.60×10^{-3}**

pruned model with SAPF outperforms the original model at an IoU threshold of 0.75. Additionally, the FLOPs and parameters are reduced by 6% and 4%, respectively, compared to the original model. Overall, across the three datasets, SAPF not only enhances feature representation and efficiency in Faster-RCNN [18] but also demonstrates its effectiveness in improving object detection performance.

In order to more intuitively demonstrate the feature representation capability of Faster-RCNN [18], this article presents the visualization results of the NWPU VHR-10 [20] dataset on Faster-RCNN [18] in Fig. 4. Among them, Fig. 4(a)–(h) shows the visualization results of inference on the NWPU VHR-10 [20] dataset on a fully trained Faster-RCNN [18]; Fig. 4(i)–(p) shows the visualization results of inference on Faster-RCNN [18] of NWPU VHR-10 [20] dataset after SAPF pruning. After comparing Fig. 4(a)–(h) with Fig. 4(i)–(p), it can be found that although SAPF simplifies the structure of Faster-RCNN [18], it does not affect the performance of the model in object detection. Therefore, the effectiveness and robustness of SAPF were verified.

2) Results and Analysis of the Ablation Experiments for YOLO v5s [19]: The ablation experiment results for YOLO v5s [19] on three object detection datasets are shown in Table II. For the NWPU VHR-10 dataset [20], although the accuracy of MSP is higher compared to SSP and SAPF, the FLOPs and parameters do not show a significant reduction. While SSP has similar accuracy to SAPF, SAPF is more lightweight and has the highest network efficiency. For the RSOD dataset [21], the accuracy of SSP, MSP, and SAPF exceeds that of the fully trained model, both at IoU@50 and across the IoU range from 50% to 95% (IoU@50:95). This indicates that both multiscale and single-scale convolutional layers contain redundant filters, which negatively impact the feature representation capability of those layers. Compared to SSP and MSP, SAPF has lower FLOPs and parameters, demonstrating higher network training efficiency. For the SIMD dataset [22], the accuracy of the four methods is essentially similar. Additionally, the pruning levels of SSP and MSP are quite comparable. SAPF not only maintains the feature representation capability of YOLO v5s [19] but also effectively reduces the computational resources required by the network, thereby enhancing model efficiency.

Similar to Faster-RCNN [18], Fig. 5 visually displays the visualization results of object detection on the NWPU VHR-10 [20] dataset on YOLO v5s [19]. Among them, Fig. 5(a)–(h) shows the visualization results of inference from the NWPU VHR-10 [20] dataset on fully-trained YOLO v5s [19]; Fig. 5(i)–(p) shows the visualization results of inference on YOLO v5s [19] of NWPU VHR-10 [20] dataset after SAPF pruning. By comparing Fig. 5(a)–(h) with Fig. 5(i)–(p), the objects recognized by fully-trained YOLO v5s [19] can still be recognized on YOLO v5s [19] after SAPF pruning. This means that YOLO v5s [19] after SAPF sparsity will still not affect the performance of the model in object detection. Therefore, the effectiveness and generalization of SAPF are further validated.

D. Parameters Sensitivity Analysis

In this part, the sensitivity of two parameters in SAPF will be analyzed: the truncation threshold Γ in SSP and the feature importance similarity β in MSP.

1) Sensitivity Analysis of the Truncation Threshold Γ: In SSP, the SVD technique maps the filters of single-scale convolutional layers from the convolutional space to the feature space to obtain eigenvalues, which are then used to measure the importance of each filter. The problem of whether a filter should be retained is evaluated by setting a truncation threshold Γ. Filters with eigenvalues greater than the truncation threshold Γ are considered important and retained, while those below the threshold are deemed unimportant and removed. This sensitivity experiment sets three different Γ thresholds and performs experiments with Faster-RCNN [18] and YOLO v5s [19] across three object detection datasets. The experimental results are shown in Fig. 6.

As shown in Fig. 6(a), both Faster-RCNN [18] and YOLO v5s [19] exhibit a gradual decrease in mAP@50 as the Γ value increases. This is because a higher Γ value corresponds to a higher pruning rate. Fig. 6(b)–(c) demonstrates that higher Γ

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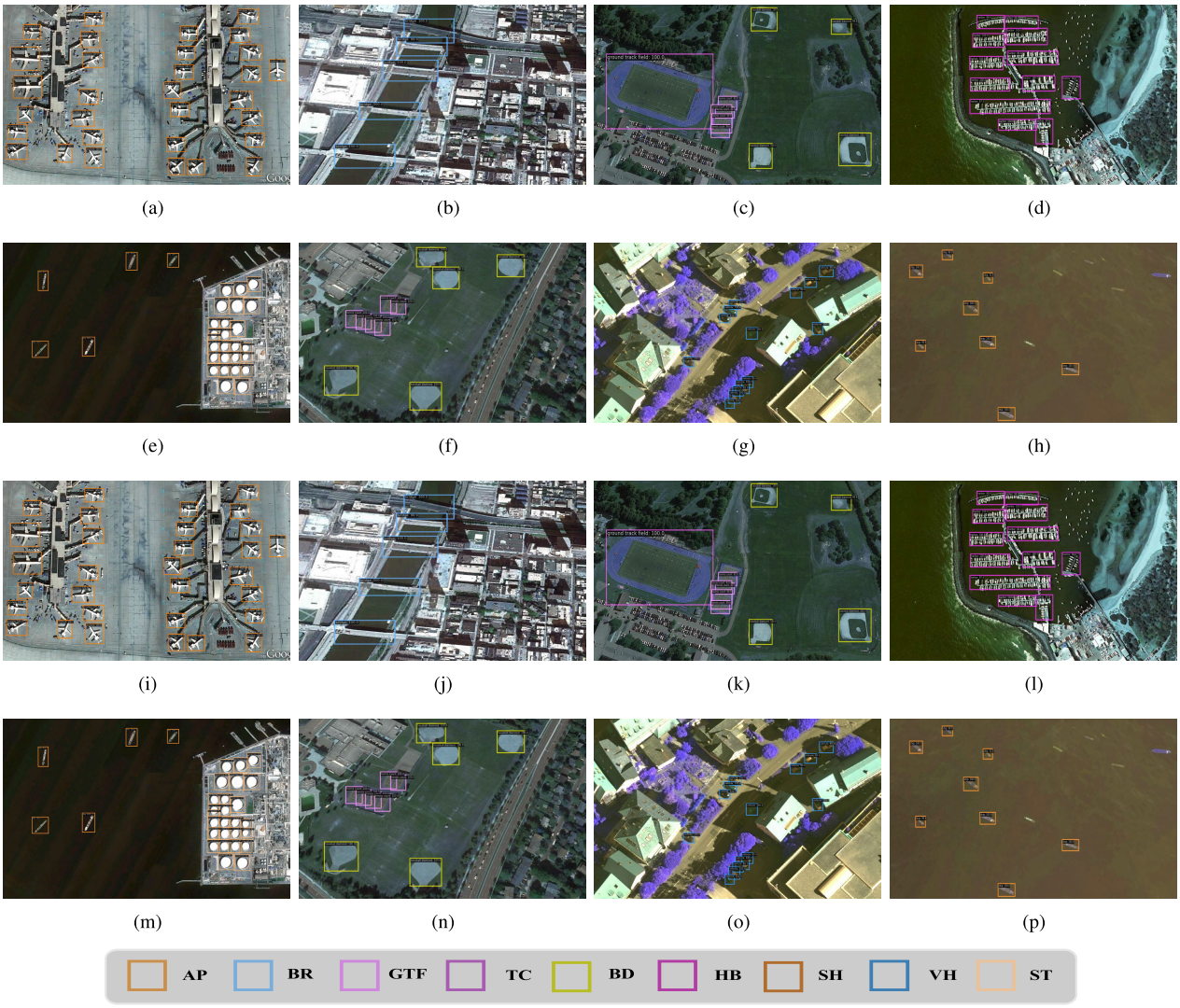


Fig. 4. Visualization results of the Faster-RCNN model on the NWPU VHR-10 dataset. (a)–(h) Visualization results of NWPU VHR-10 inference after complete training on Faster-RCNN. (i)–(p) Visualization results of Faster-RCNN inference of NWPU VHR-10 after SAPF pruning. AP, BD, BC, SH, ST, TC, VH, and HB, respectively, represent airplanes, baseball diamonds, basketball courts, ships, storage tanks, tennis courts, vehicles, and harbors.

TABLE II

ABLATION EXPERIMENT RESULTS FOR THE YOLO V5S [19] MODEL. THE BOLDED PARTS ARE THE SAPF METHOD AND ITS RELATED METRIC RESULTS

Model	Datasets	SSP	MSP	mAP@50 (%)	mAP@(50:95) (%)	FLOPs (G)	Parameters (M)	Efficiency (GPUDays/Epochs)
YOLO v5s [19]	NWPU VHR-10 [20]			90.60	59.30	8.31	7.24	2.51×10^{-4}
		✓		89.60	57.00	7.84	6.05	1.09×10^{-4}
			✓	90.40	57.80	7.99	6.93	1.30×10^{-4}
		✓	✓	**89.60**	**57.10**	**7.52**	**5.74**	$\mathbf{0.95 \times 10^{-4}}$
	RSOD [21]			86.30	56.50	8.31	7.23	1.25×10^{-4}
		✓		82.20	61.30	7.84	6.05	1.19×10^{-4}
			✓	90.12	63.20	7.99	6.93	1.17×10^{-4}
		✓	✓	**90.20**	**63.80**	**7.52**	**5.74**	$\mathbf{1.10 \times 10^{-4}}$
	SIMD [22]			80.00	65.00	8.31	7.24	7.67×10^{-4}
		✓		79.60	63.30	7.89	6.82	6.48×10^{-4}
			✓	79.30	64.20	7.89	6.58	6.19×10^{-4}
		✓	✓	**79.50**	**65.20**	**7.83**	**6.52**	$\mathbf{5.96 \times 10^{-4}}$

values result in smaller FLOPs and parameters. This indicates that as Γ increases, the network structure becomes sparser, thereby improving network efficiency. Within this range of thresholds, a Γ value of 0.4 is found to strike a balance between minimal accuracy loss and significant improvement in network efficiency for both models. Therefore, Γ is set to 0.4 as the optimal truncation threshold in the SSP module of SAPF for pruning single-scale convolutional layers.

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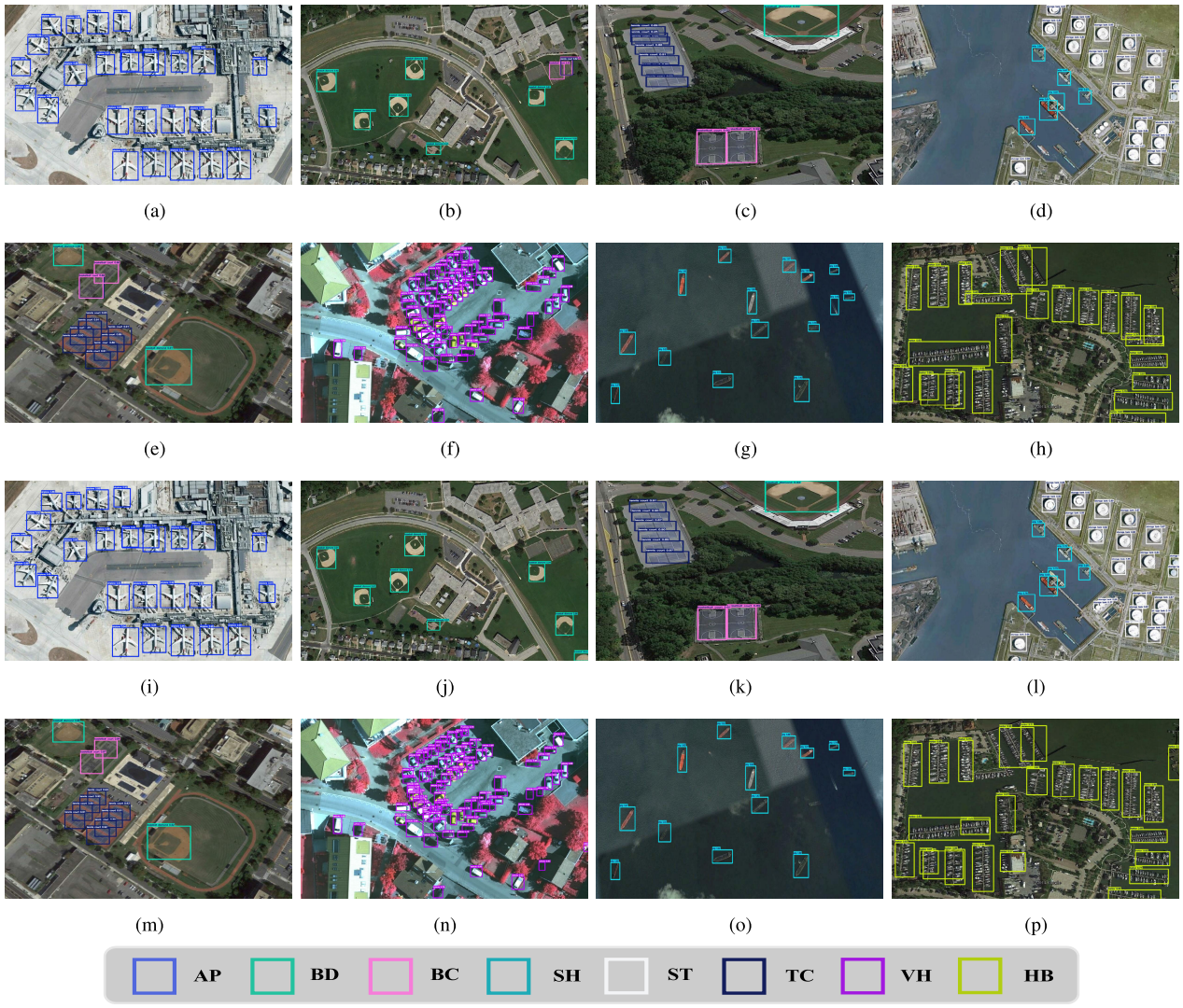


Fig. 5. Visualization results of the YOLO v5s model on the NWPU VHR-10 dataset. (a)–(h) Visualization results of NWPU VHR-10 inference after complete training on YOLO v5s; (i)–(p) Visualization results of YOLO v5s inference of NWPU VHR-10 after SAPF pruning. AP, BR, GTF, TC, BD, HB, SH, VH, and ST, respectively, represent airplanes, bridges, ground track fields, tennis courts, baseball diamonds, harbors, ships, vehicles, and storage tanks.

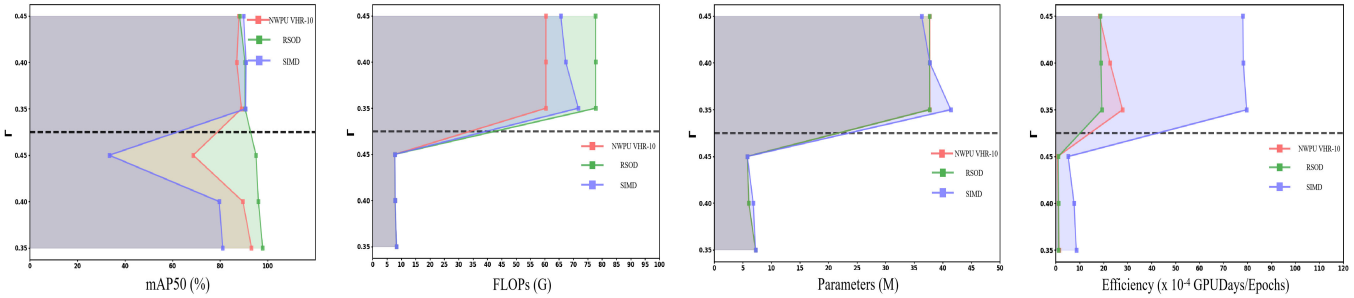
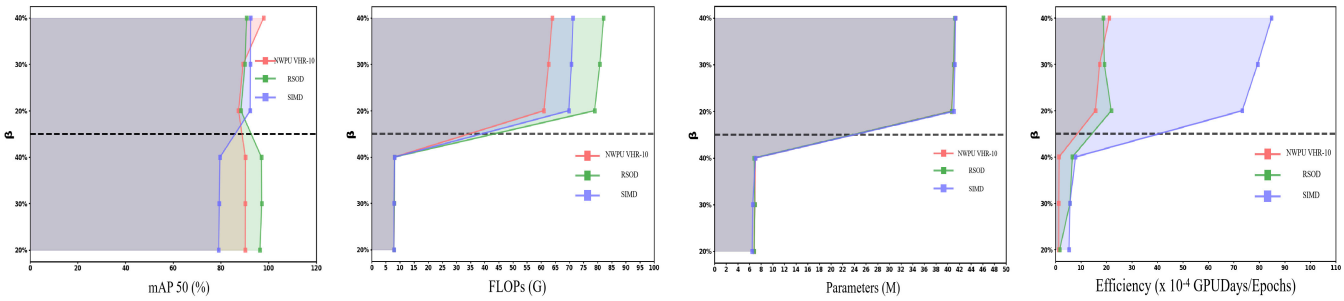


Fig. 6. Sensitivity results of the truncation threshold Γ. (a) Measures the accuracy of Faster-RCNN and YOLO v5s with mAP@50. (b) and (c) Calculate the FLOPs and parameters of the models, respectively. (d) Assesses the efficiency. The upper half of each plot shows the results related to Faster-RCNN, while the lower half displays the results related to YOLO v5s.

2) Sensitivity Analysis of the Feature Importance Similarity β: In MSP, SIS is used to measure the information richness between multiscale convolutional layers, thereby assessing the redundancy of filters across different scales. This study employs an importance sensitivity parameter β to determine the global importance of filters in multiscale convolutional layers. Specifically, a larger β value indicates higher importance for that scale of the convolutional layer, resulting in a lower

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(a) (b) (c) (d)

Fig. 7. Sensitivity results of the feature importance similarity β. (a) Measures the accuracy of Faster-RCNN and YOLO v5s with mAP@50. (b) and (c) Calculate the FLOPs and parameters of the models, respectively. (d) Assess the efficiency. The upper half of each plot shows the results related to Faster-RCNN, while the lower half displays the results related to YOLO v5s.

TABLE III

COMPARISONS OF SAPF AND OTHER NETWORK COMPRESSION METHODS ON FASTER-RCNN. THE BOLDED PARTS ARE THE SAPF METHOD AND ITS RELATED METRIC RESULTS

Model	Datasets	Methods	mAP@50 (%)	mAP@75 (%)	FLOPs (G)	Parameters (M)	Efficiency (GPUDays/Epochs)
Faster-RCNN [18]	NWPU VHR-10 [20]	L1-Norm [31]	87.00	59.10	60.59	38.93	1.79×10^{-3}
		Taylor Pruning [36]	88.12	56.62	54.28	38.31	1.64×10^{-3}
		EFPF [37]	88.40	58.50	59.41	37.75	1.78×10^{-3}
		Network Slimming [35]	87.60	55.50	59.90	38.33	1.66×10^{-3}
		Soft Elastic Pruning [38]	88.04	57.02	51.01	37.89	0.88×10^{-3}
		ThiNet [39]	85.65	57.41	59.02	39.20	1.81×10^{-3}
		EDropout [40]	85.64	57.39	54.38	37.39	0.89×10^{-3}
		DCP [41]	85.54	56.85	58.46	37.90	0.94×10^{-3}
		SAPF	**89.40**	**62.00**	**60.34**	**37.36**	$\mathbf{0.78 \times 10^{-3}}$
	RSOD [21]	L1-Norm [31]	88.10	69.00	75.85	38.89	1.31×10^{-3}
		Taylor Pruning [36]	88.12	67.69	69.97	39.36	1.32×10^{-3}
		EFPF [37]	87.50	67.30	77.74	37.71	1.37×10^{-3}
		Network Slimming [35]	88.50	67.90	74.45	38.30	1.29×10^{-3}
		Soft Elastic Pruning [38]	88.03	67.63	68.34	38.48	1.28×10^{-3}
		ThiNet [39]	88.75	68.54	70.47	39.55	1.33×10^{-3}
		EDropout [40]	88.69	68.50	64.93	37.72	1.28×10^{-3}
		DCP [41]	88.32	68.52	69.81	38.24	1.30×10^{-3}
		SAPF	**90.40**	**68.80**	**75.02**	**37.35**	$\mathbf{1.26 \times 10^{-3}}$
	SIMD [22]	L1-Norm [31]	92.30	92.20	66.24	38.95	8.33×10^{-3}
		Taylor Pruning [36]	91.28	89.63	53.00	37.43	7.70×10^{-3}
		EFPF [37]	90.90	88.80	67.33	37.76	7.33×10^{-3}
		Network Slimming [35]	91.40	90.20	59.18	38.33	7.80×10^{-3}
		Soft Elastic Pruning [38]	91.19	89.54	50.00	38.25	7.45×10^{-3}
		ThiNet [39]	92.19	91.20	59.83	38.89	8.29×10^{-3}
		EDropout [40]	92.17	91.21	55.13	36.60	7.71×10^{-3}
		DCP [41]	92.08	91.17	59.27	37.12	7.75×10^{-3}
		SAPF	**92.10**	**91.70**	**65.52**	**37.43**	$\mathbf{7.60 \times 10^{-3}}$

pruning rate. Conversely, a smaller β value signifies higher redundancy in the filters of that scale of the convolutional layer, leading to a higher pruning rate.

As shown in Fig. 7(a), the highest accuracy is achieved when β is set to 40%. Although mAP@50 gradually decreases as β decreases, both FLOPs and parameters also decrease progressively in Fig. 7(b)–(c). This indicates that β determines the global importance of multiscale convolutional layers. This means that different object detection tasks can adjust the importance of the network structure based on the setting of β. By tuning β, the sparsity of the model structure can be tailored to meet the specific requirements of different object detection tasks. As shown in Fig. 7(d), when β is 30%, the efficiency is also objective. Based on a comprehensive evaluation, this study selects β at 30% as the optimal value for the sparsity of multiscale convolutional layers.

E. Comparisons With Other Pruning Methods

In this section, the effectiveness of the proposed pruning framework is validated by comparing SAPF with other existing pruning methods. We selected eight popular pruning methods as SOTA experiments for comparative analysis with

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TABLE IV
COMPARISONS OF SAPF AND OTHER NETWORK COMPRESSION METHODS ON YOLO V5S. THE BOLDED PARTS ARE THE SAPF METHOD AND ITS RELATED METRIC RESULTS

Model	Datasets	Methods	mAP@50 (%)	mAP@(50:95) (%)	FLOPs (G)	Parameters (M)	Efficiency (GPUDays/Epochs)
YOLO v5s [19]	NWPU VHR-10 [20]	L1-Norm [31]	81.10	47.90	6.71	5.93	1.34×10^{-4}
		Taylor Pruning [36]	71.39	38.77	6.22	5.60	1.15×10^{-4}
		EFPF [37]	77.00	47.90	7.17	4.73	0.89×10^{-4}
		Network Slimming [35]	51.60	29.50	6.35	5.61	1.25×10^{-4}
		Soft Elastic Pruning [38]	71.32	38.73	5.17	5.24	1.30×10^{-4}
		ThiNet [39]	68.45	38.96	6.68	5.81	1.28×10^{-4}
		EDropout [40]	68.42	38.97	6.15	5.54	1.01×10^{-4}
		DCP [41]	67.39	38.62	6.61	5.62	1.21×10^{-4}
		SAPF	**89.60**	**57.10**	**7.52**	**5.74**	$\mathbf{0.95 \times 10^{-4}}$
	RSOD [21]	L1-Norm [31]	90.00	62.20	7.39	6.65	1.32×10^{-4}
		Taylor Pruning [36]	90.20	65.20	6.22	5.60	1.27×10^{-4}
		EFPF [37]	87.50	61.70	6.17	4.72	1.01×10^{-4}
		Network Slimming [35]	90.00	65.30	6.35	5.61	1.27×10^{-4}
		Soft Elastic Pruning [38]	90.60	64.90	5.17	5.24	1.15×10^{-4}
		ThiNet [39]	89.93	64.61	6.71	6.04	1.29×10^{-4}
		EDropout [40]	89.81	64.53	6.19	5.76	1.27×10^{-4}
		DCP [41]	89.76	64.48	6.65	5.84	1.29×10^{-4}
		SAPF	**90.20**	**63.80**	**7.52**	**5.74**	$\mathbf{1.10 \times 10^{-4}}$
	SIMD [22]	L1-Norm [31]	76.50	58.00	7.39	6.65	7.02×10^{-4}
		Taylor Pruning[36]	72.19	54.70	5.89	6.83	6.25×10^{-4}
		EFPF [37]	79.60	63.30	7.89	6.82	6.49×10^{-4}
		Network Slimming [35]	60.80	45.90	7.11	6.46	6.35×10^{-4}
		Soft Elastic Pruning [38]	70.24	54.64	5.93	6.68	6.45×10^{-4}
		ThiNet [39]	70.25	53.19	6.84	6.77	6.95×10^{-4}
		EDropout [40]	70.17	53.16	6.30	6.46	6.21×10^{-4}
		DCP [41]	70.16	53.09	6.77	6.54	6.25×10^{-4}
		SAPF	**79.50**	**65.20**	**7.83**	**6.52**	$\mathbf{5.96 \times 10^{-4}}$

SAPF, including traditional pruning methods such as **"L1-Norm"** pruning [31], **"Taylor pruning"** [36], **"EFPF"** [37], **"Network Slimming"** [35], **"Soft Elastic Pruning"** [38], **"ThiNet"** [39], **"EDropout"** [40], and **"DCP"** [41]. The results of applying these pruning methods and SAPF on Faster-RCNN [18] across three object detection datasets are presented in Table III. The results of applying these pruning methods and SAPF on YOLO v5s [19] across the same three object detection datasets are shown in Table IV. These comparisons allow for the assessment of the performance of different pruning methods in remote sensing object detection practical applications and the verification of the effectiveness of SAPF.

From Table III, it can be seen that the application of SAPF to the Faster-RCNN [18] demonstrates significant superiority. By comparing different pruning methods across various remote sensing object detection datasets, SAPF shows advantages in multiple key performance metrics. First, SAPF typically achieves the best performance in the core metrics mAP@50 and mAP@75, indicating that the model pruned with SAPF exhibits superior feature representation ability compared to other pruning methods. Second, although SAPF does not achieve the lowest FLOPs and parameters on certain datasets, the efficiency of the model pruned with SAPF has been significantly improved. This means that SAPF allows the model to achieve better performance with fewer computational resources and less training time, which is important for optimizing computational resources and controlling training time costs in practical remote sensing object detection applications. Moreover, whether on the high-resolution NWPU VHR-10 dataset [20] or the more challenging RSOD dataset [21] and SIMD dataset [22], SAPF maintains stable performance. On the NWPU VHR-10 [20] and RSOD [21] datasets, the model pruned with SAPF achieves the best object detection performance, while also demonstrating notable improvements in terms of parameter count and model efficiency. For the SIMD dataset [22], although the performance is slightly lower than **"L1-Norm"** pruning [31], SAPF still achieves a significant reduction in model complexity. Overall, this cross-dataset stability indicates that SAPF can adapt to various remote sensing object detection application scenarios and data characteristics, further validating the robustness and generalization capability of SAPF.

Similar to the Faster-RCNN [18], Table IV shows that SAPF exhibits high detection results in YOLO v5s [19] for targets on all three remote sensing datasets, which implies that SAPF has strong robustness in object detection. It is noteworthy that on the NWPU VHR-10 dataset [20], while pruning methods such as **"EDropout"** [40], **"Soft Elastic Pruning"** [38], and **"DCP"** [41] reduce model complexity, the target detection performance of SAPF is approximately 20% higher than these methods. Additionally, SAPF achieves the highest

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performance on the RSOD dataset [21]. Although it is comparable to the **"Soft Elastic Pruning"** [38] method in terms of performance, SAPF offers lower model complexity and higher computational efficiency. On the SIMD dataset [22], the accuracy of SAPF is second only to that of **"L1-Norm"** [31], which means that while other traditional pruning methods achieve excellent pruning rates on SIMD [22], the representation capability of the pruned models is poor. Although **"L1-Norm"** [31] achieves slightly higher accuracy than SAPF, the pruning capability of **"L1-Norm"** [31] is inferior. In summary, SAPF not only maintains the feature information of the model but also effectively performs pruning.

By analyzing the comparative experiments in Tables III and IV, it can be concluded that both on Faster-RCNN [18] and YOLO v5s [19], the SAPF pruned model shows high performance in target detection and effectively reduces the complexity of the model, thus accelerating the training efficiency of the model. This means that SAPF can accurately remove the redundant filters to enhance the feature representation of the model. This also validates the generalization and robustness of SAPF for remote sensing target detection applications.

V. Conclusion

In this article, an SAPF is proposed to save computational resources and enhance the efficiency of object detection models by leveraging multifeature representations for object detection tasks. For single-scale convolutional layers with unique scaling factors, the importance of the feature representation of the filter is measured by mapping the filter from the convolutional space to the feature space using SVD techniques. This approach is employed to assess the redundancy of single-scale convolutional layers and perform pruning. Additionally, for multiscale convolutional layers with various scaling factors, an SIS is designed to evaluate the sensitivity of feature information richness between different scales. Through SIS, the feature representation capabilities of convolutional layers at different scales are assessed to ensure that filters with a negative impact on object detection, while preserving critical information, are precisely pruned. By using SAPF to detect objects on the NWPU VHR-10 [20], RSOD [21], and SIMD [22] datasets in Faster-RCNN [18] and YOLO v5s [19] models, it was observed that the performance of the pruned models was not significantly affected, with a significant reduction in FLOPs and parameters. The training efficiency of the model can be doubled at most. Compared with other traditional pruning methods, SAPF seems to have more generalization ability, and the detection results of the object after SAPF pruning are usually 1%–2% higher than other methods. This means that SAPF accurately prunes redundant filters on Faster-RCNN [18] and YOLO v5s [19], maximizing the structure of sparse models while ensuring performance and improving model efficiency. Based on this, the effectiveness and generalization of SAPF have been demonstrated. SAPF is capable of performing more precise and in-depth pruning of object detection tasks while ensuring the accuracy of the object detection model.

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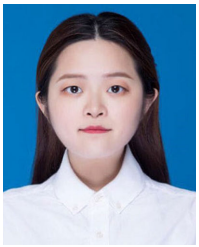
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